CCC-C-442D
April 5, 1974
SUPERSEDING
Fed. Spec. CCC-C-442C
April 3, 1969

FEDERAL SPECIFICATION

CLOTH, DUCK, COTTON, BLEACHED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers bleached cotton duck utilized in the manufacturer of clothing items (see 6.1).

1.2 Classification. The bleached cotton duck shall be of the following types as specified (see 6.1 and 6.2).

Type I	-	5.0 ounces per square yard.
Type II	-	7.0 ounces per square yard.
Type III	-	8.2 ounces per square yard.
Type IV	-	8.5 ounces per square yard.
Type V	-	7.0 ounces per square yard.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

V-T-276	-	Thread, Cotton.
PPP-P-1135	-	Packaging and Packing of Duck Fabrics (Cotton: Synthetic Fiber: Cotton - Synthetic Fiber Blends).

Federal Standards:

Fed. Std. No. 4	-	Glossary of Fabric Imperfections.
Fed. Std. No. 191	-	Textile Test Methods.

FSC 8305

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

PUBLICATION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is required, the cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Yarn. The yarn shall be made from cotton which has been carded, drawn, and twisted into single or plied yarns as specified in table I.

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3.3 Color. The cloth shall be chemically bleached white and may be supplemented with optical brighteners which fluoresce to the blue region. The degree of whiteness shall match the standard sample. The finished bleached cloth shall not discolor to a degree greater than that shown by the standard sample when tested as specified in 4.5. When no standard sample is available, the discoloration shall be no lower than the rating as "Fair" when tested as specified in 4.5.

3.3.1 Matching. The color shall match the available standard sample (see 6.3) under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.4 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.5.

TABLE I - Physical requirements

Type	Weight per square yard minimum	Yarn ply		Yarns per inch minimum		Breaking strength minimum	
		Warp	Filling	Warp	Filling	Warp	Filling
	Ounces					Pounds	Pounds
I	5.0	2	2	77	44	130	75
II	7.0	1	2	88	26	110	60
III	8.2	2	2	58	48	135	100
IV	8.5	1	4	112	32	120	120
V	7.0	2	2	52	34	80	48

3.4.1 Width. The width shall be as specified (see 6.2) and shall be the minimum acceptable width, inclusive of selvages.

3.5 Weave. The weave for types I, III and V cloths shall be plain, one up, one down. The weave for types II and IV cloths shall be an oxford weave with two ends weaving as one and alternating in plain-weave order with each pick.

3.6 Finish. The cloth shall be desized, singed, and scoured, bleached and mercerized.

3.6.1 Nonfibrous materials. The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed 2.0 percent when tested as specified in 4.5.

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3.7 Shrinkage. The cloth shall be preshrunk so that the residual shrinkage shall be not more than 2.0 percent in either the warp or filling direction when tested as specified in 4.5. The preshrinking process used shall not be identified in any manner on the cloth, the ticket, or on the package.

3.8 pH. The pH value of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.5.

3.9 Seam efficiency. The seam efficiency of the finished cloth shall be not less than 80 percent when tested as specified in 4.5.

3.10 Length and put-up. Unless otherwise specified (see 6.2, the cloth shall be in continuous lengths, each not less than 50 yards. The cloth shall be put-up on rolls as specified in PPP-P-1135.

3.11 Face identification. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the roll.

3.12 Fiber identification. Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.13 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the applicable established maximum point values.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

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4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended or qualified in this specification or applicable purchase document.

4.3 Examination of the end item. Examination of the end item shall be in accordance with 4.3.1 through 4.3.3.

4.3.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. All defects, as defined in section I of Fed. Std. No. 4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.3.1.1 except that only those slubs and knots which exceed the limits shown in figure 1 of Fed. Std. No. 4 shall be scored. No linear yard (increments of 1 yard on the measuring device or the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 35 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 53 points. If one roll exceeds 53 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 53 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.3.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	-	one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	-	two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	-	three points
For defects exceeding 9 inches in any dimension	-	four points

The following defects when present shall be scored four points for each yard in which they occur:

Baggy, ridgy or wavy cloth.
Width less than specified.
Overall uncleanness.
Objectionable odor.

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4.3.2 Examination for length.

4.3.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.3.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.3.3 Examination for non-identification of preshrinking process, face identification and compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination, each piece in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample.

Face identification missing from either or both ends.
Face identification on wrong side.
Preshrinkage process identified by name or trademark on cloth or ticket.
Not labeled, ticketed, or invoiced in accordance with the Rules and Regulations Under the Textiles Fiber Products Identification Act.

4.4 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1135 to determine whether packaging, packing, and marking comply with section 5 requirements.

4.5 Testing of the end item. The methods of testing specified in Fed. Std. No. 191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3 continuous yards, full width of the finished cloth. The sample size shall be as listed in table II. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more units fail to meet any requirements specified. All test reports shall contain the individual values utilized in expressing the final results.

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TABLE II - Sample size

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III - Test methods

Characteristic	Requirement paragraph	Test method
Discoloration	3.3	5660 1/ 2/
Weight	3.4	5041
Yarn ply:		
Warp	3.4	Visual 2/
Filling	3.4	Visual 2/
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Weave	3.5	Visual 2/
Singed	3.6	3/
Mercerization	3.6	Microscopic examination for swelling of cotton fiber 2/
Nonfibrous material	3.6.1	2611
Shrinkage in laundering	3.7	5550

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TABLE III - Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
pH	3.8	2811
Seam efficiency	3.9	5110 4/

1/ Method 5660 shall be used, except that exposure of the specimen and standard sample shall be for 20 standard fading hours. Any specimen showing discoloration less than or equal to that of the standard sample shall be rated "pass". Any specimen showing discoloration greater than that of the standard sample shall be rated "fail". When no standard sample is available, the discoloration shall be considered satisfactory if it is not worse than the rating of "fair" in the scale indicated in Method 5660.

2/ One determination for each sample unit and the result reported as "pass" or "fail".

3/ Unless otherwise specified, a certificate of compliance is required and will be acceptable for the requirement stated.

4/ The thread shall be cotton conforming to type IA2 of specification V-T-276. The needle thread shall be ticket No. 24, 3 ply and the looper thread shall be ticket No. 60, 3 ply. The needle size shall measure 0.049 $\pm$ 0.001 inch across the blade at the eye.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1135.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1135.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1135.

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6. NOTES

6.1 Intended use. Types I, II and III bleached cotton ducks covered by this specification are intended for use in white uniforms for personnel of all Government departments. Type IV bleached cotton duck is intended for use in dress uniform trousers of U.S. Military Academy Cadets. Type V bleached cotton duck is utilized as the base cloth for Smock, Mens's, Dental, Operating; and Smock, Men's, Medical Assistant.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type and nomenclature of cloth required (see 1.2).
(c) Width required (see 3.4.1).
(d) Length of pieces if other than specified (see 3.9).
(e) Selection of applicable levels of put-up, packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Level B packaging and packing should be used by ordering agencies only when shipment is known to be made to a holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

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Air Force - 82

Review activities:

Army - MD
Navy - MS

User activity:

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GSA-FSS
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NAME OF ORGANIZATION AND ADDRESS

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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